



HIGHER MOBILITY AMONG RESEARCHERS IDENTIFIED AS POTENTIAL ACADEMIC TALENTS COMPARED TO A CONTROL GROUP

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WHAT HAS BEEN DONE

The literature on scientific mobility roughly focuses on two areas:

- 1) the experience of mobility and its properties and
- 2) the by-products and positive or negative impacts of mobility.

Performance and impact of mobile scientists or the so-called “mover’s advantage”

(Franzoni et al., 2014)

Few studies on the downsides of mobility such as instability of collaboration and gendered nature of negotiations and career decision (even between academic couples)

(Zhao et al., 2020, Schaer et al. 2017)

ENVIRONMENT VS. INDIVIDUAL'S POTENTIAL

- Performance of researchers is influenced by the institutional environment (Bonaccorsi, Belingheri, & Secondi, 2021; Li, Aste, Caccioli, & Livan, 2019)
- If environment is located in prestigious departments or universities (Amara, Landry, & Halilem, 2015; Way, Morgan, Larremore, & Clauset, 2019)
- Environment's effect is not only be of a motivational nature, but also affects collaborations such as the case of collaboration with Nobel Laurates (Li, Yin, Fortunato, & Wang, 2020; Zalewska-Kurek, Geurts, & Roosendaal, 2010)
- Individual academic ability to do research (Sugimoto et al., 2017; Franzoni et al., 2014), since environments attract and hire “the best and the brightest” (Kapur & McHale, 2005)

HAUNSCHILD AND BORNMANN (2023)

Potentially talented researchers can be identified by three basic indicators and their combinations:

- (1) the total number of papers (O),
- (2) the number of papers in high-impact (Q1) journals, and
- (3) the number of papers as a corresponding author (C)

In comparison to citation-based indicators, these take less time to accrue

The combination $O \times Q1$ performed best

DATA

For mobility

28+ million Articles and reviews; 8+ million authors; 2020 snapshot of Elsevier's Scopus database provided by Kompetenznetzwerk Bibliometrie

Mobility throughout one's career; using mode-based method to reduce noise; internal, international, both.

For talents' identification

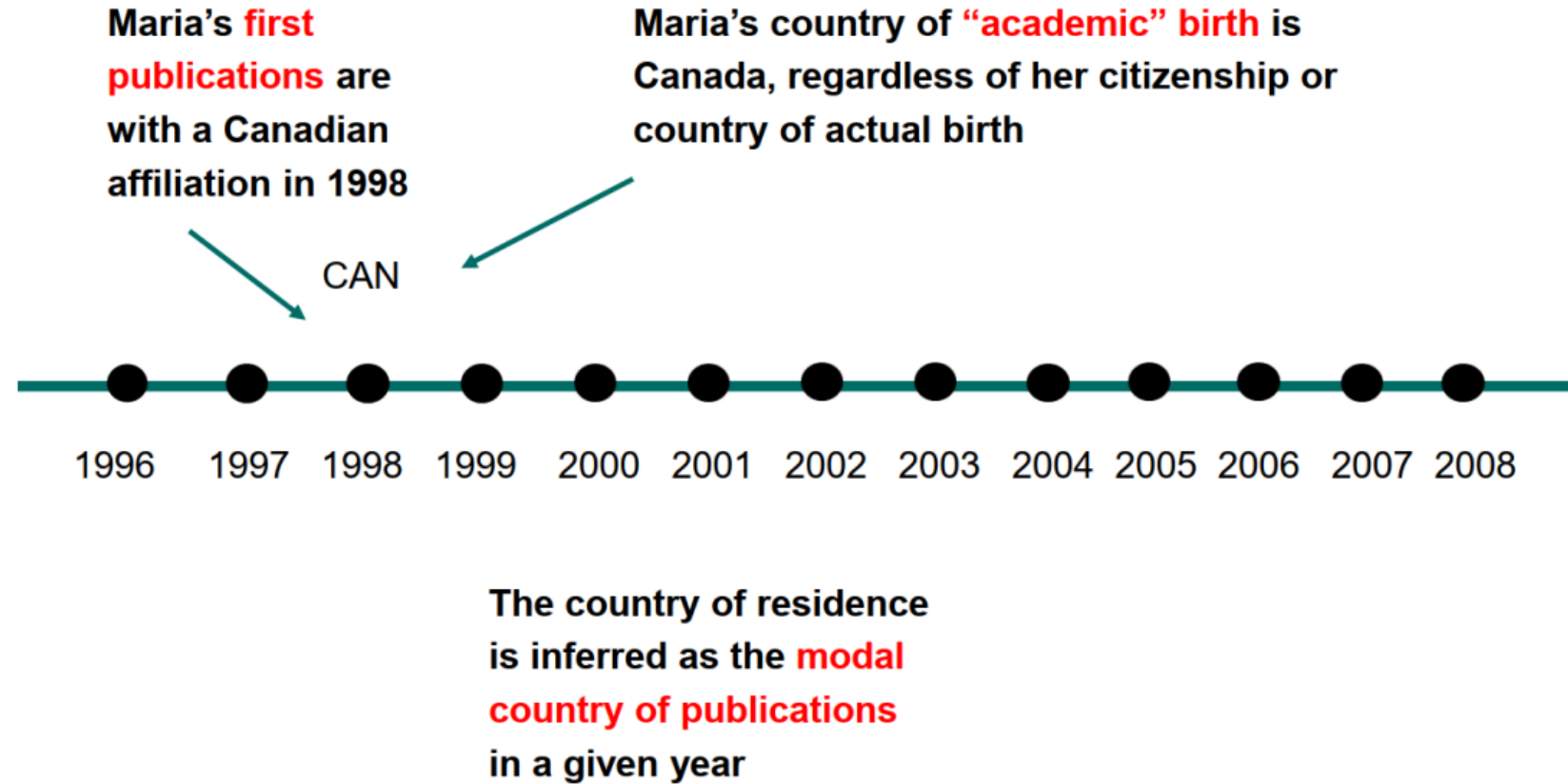
First paper in Scopus between 1999 and 2003; divided to top 1% (observation group), versus top 5 to 10% (control group)

Multinomial probit regression to find predicted probability of mobility categories

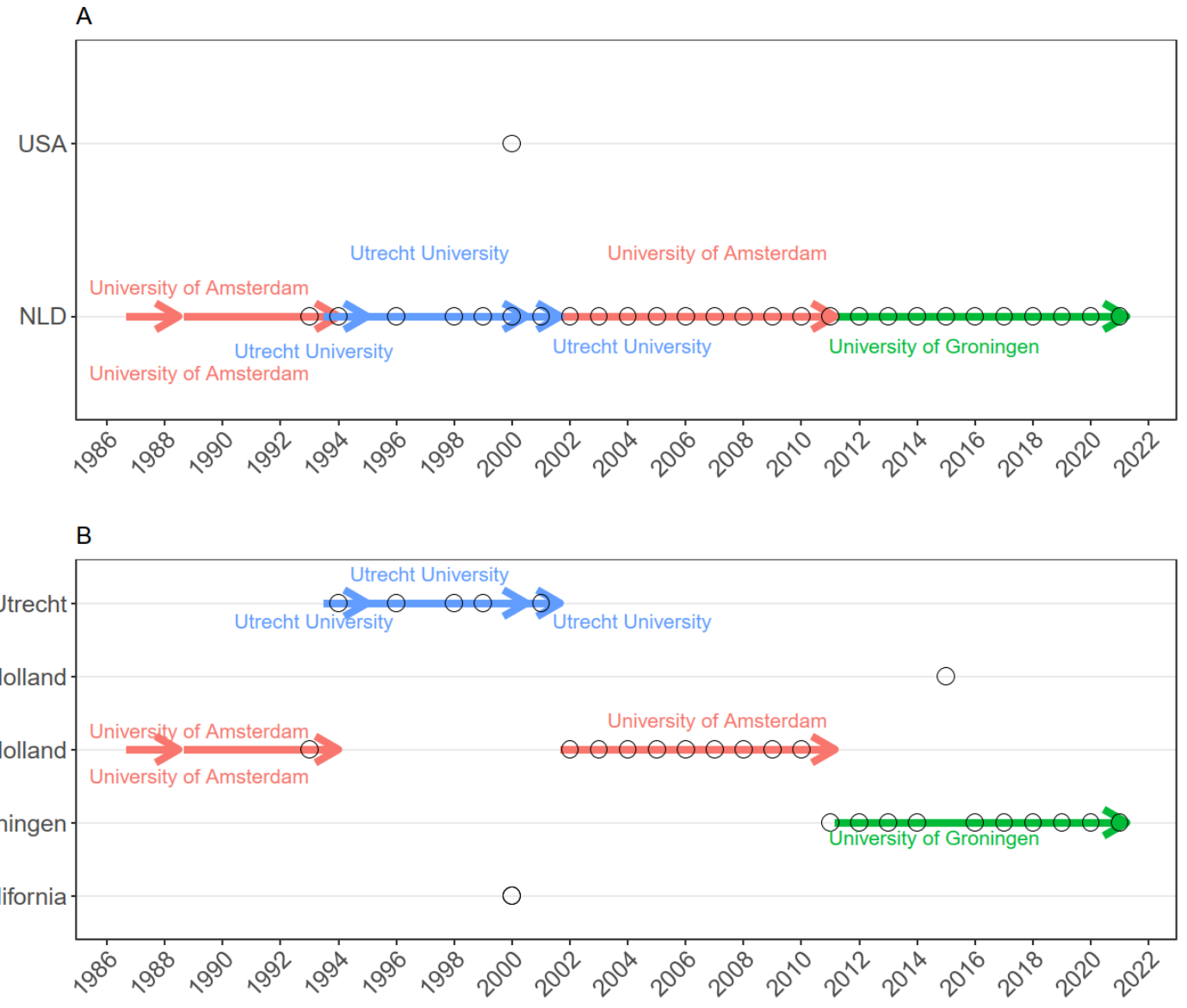


Group	Indicator combination	Number of authors before adding mobility status	Number of authors after adding mobility status
Observation group	O	41,062	39,458
	Q1	36,869	35,766
	C	39,026	37,256
	OxQ1	15,188	14,825
	OxC	18,440	17,732
	Q1xC	10,279	10,059
	OxQ1xC	7,060	6,919
Control group	O	164,176	152,677
	Q1	71,015	67,082
	C	107,179	99,110
	OxQ1	14,904	14,223
	OxC	28,804	27,013
	Q1xC	10,159	9,733
	OxQ1xC	2,980	2,864
Differences between the observation group and control group	O	-123,114	-113,219
	Q1	-34,146	-31,316
	C	-68,153	-61,854
	OxQ1	284	602
	OxC	-10,364	-9,281
	Q1xC	120	326
	OxQ1xC	4,080	4,055

MODE-BASED IDENTIFICATION OF MOBILITY

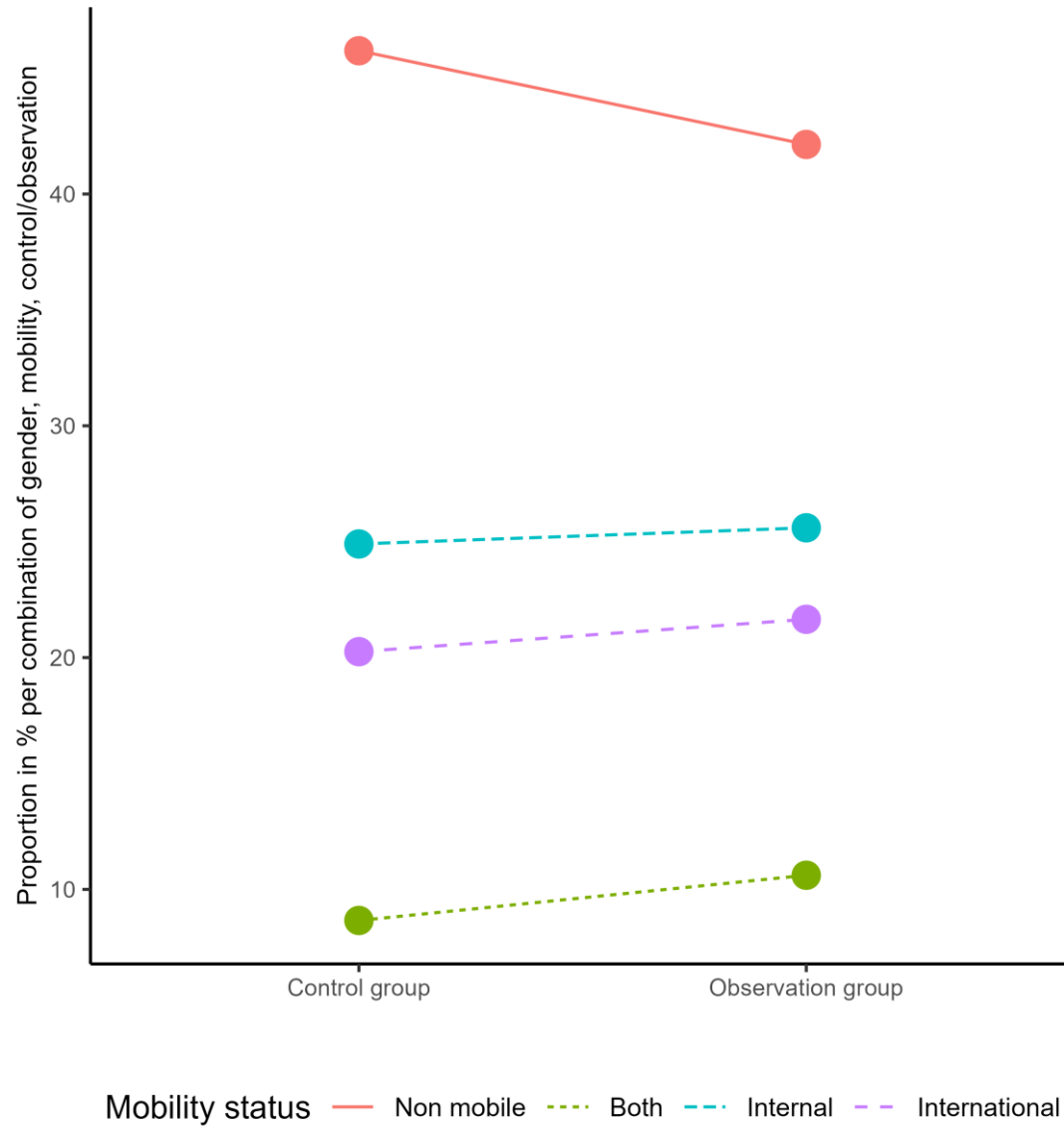


INTERNAL VERSUS INTERNATIONAL SCIENTIFIC MOBILITY; AN EXAMPLE

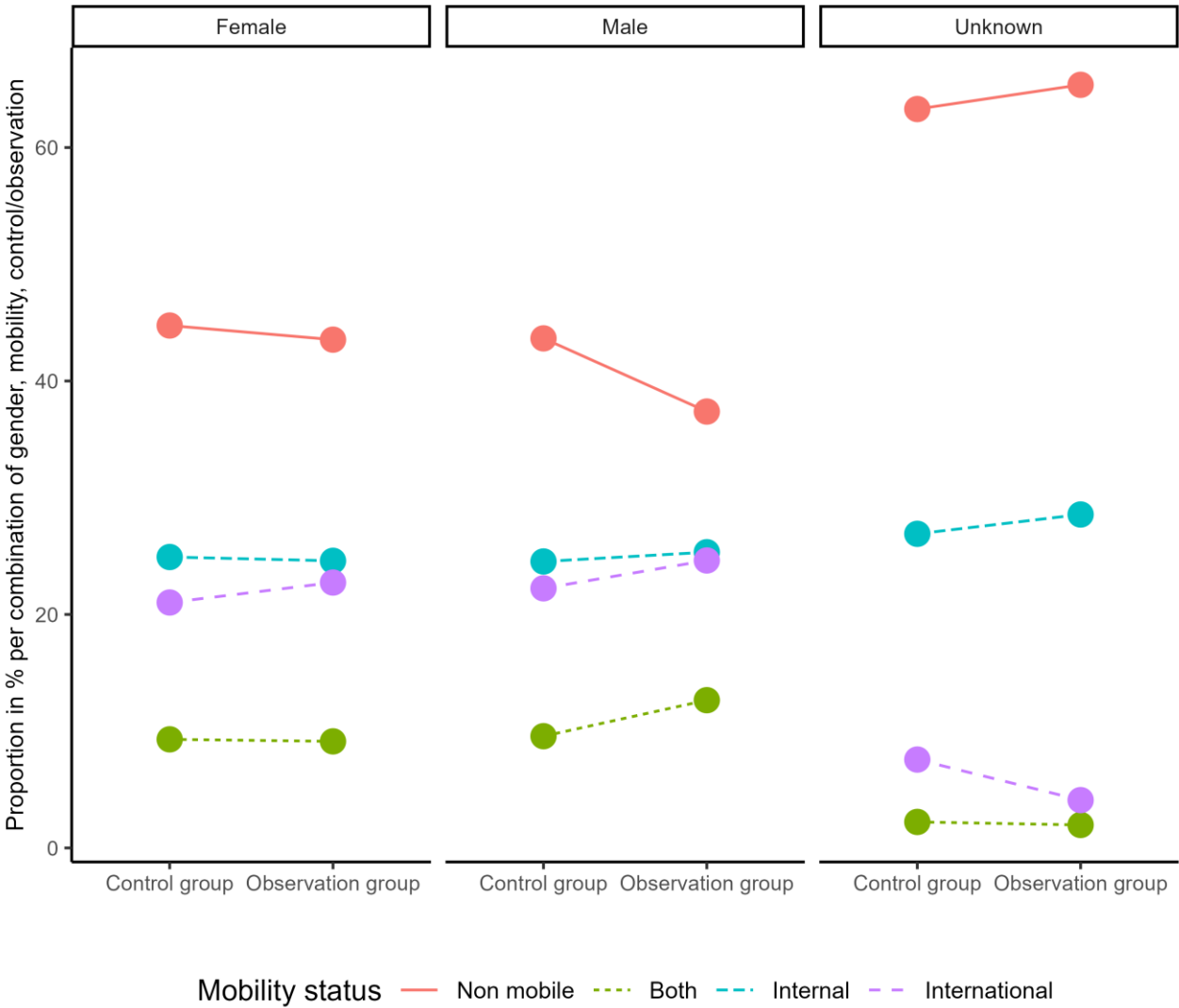
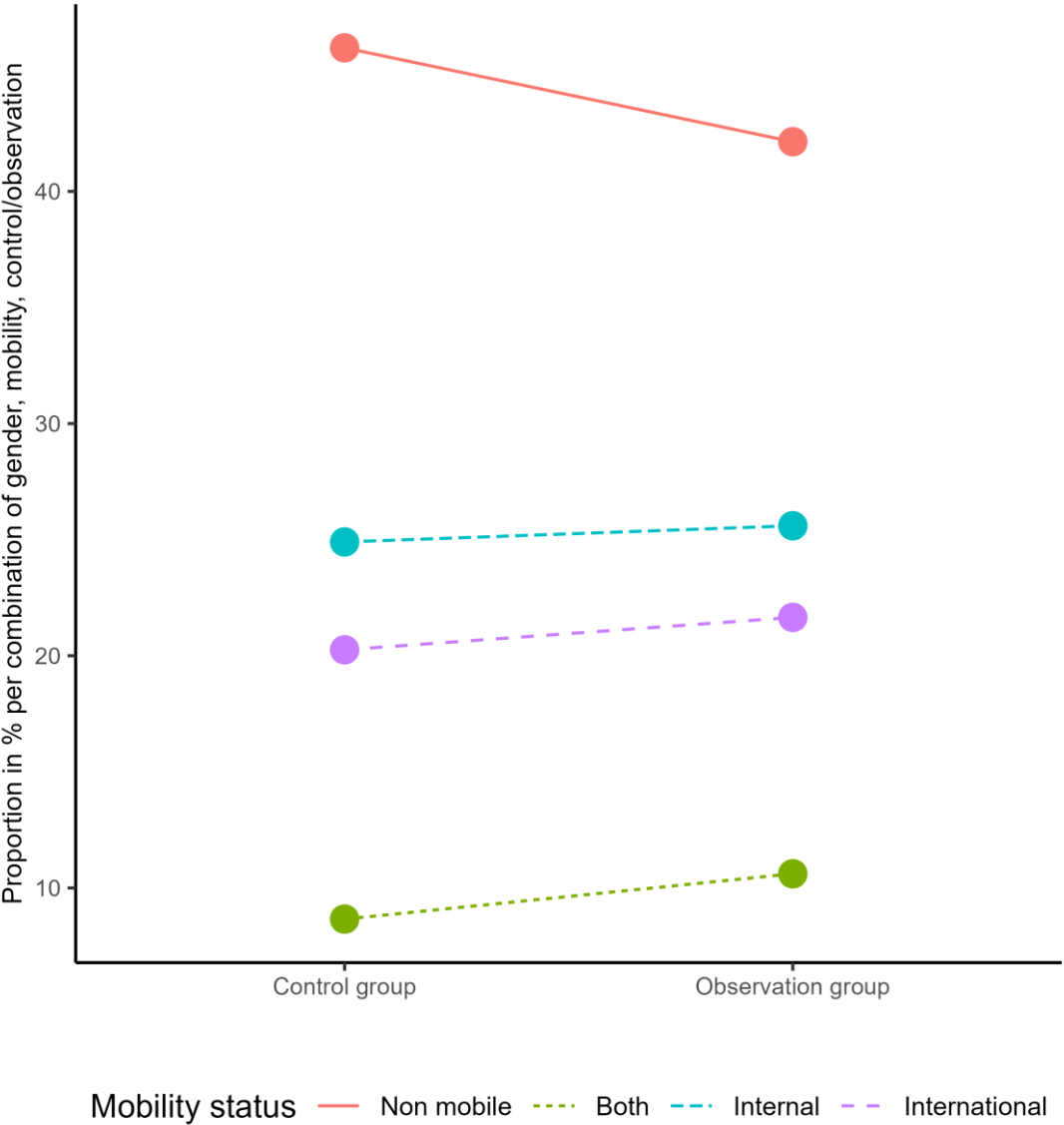


From: Akbaritabar, Danko, Zhao, Zagheni:
<https://doi.org/10.4054/MPIDR-WP-2023-038>

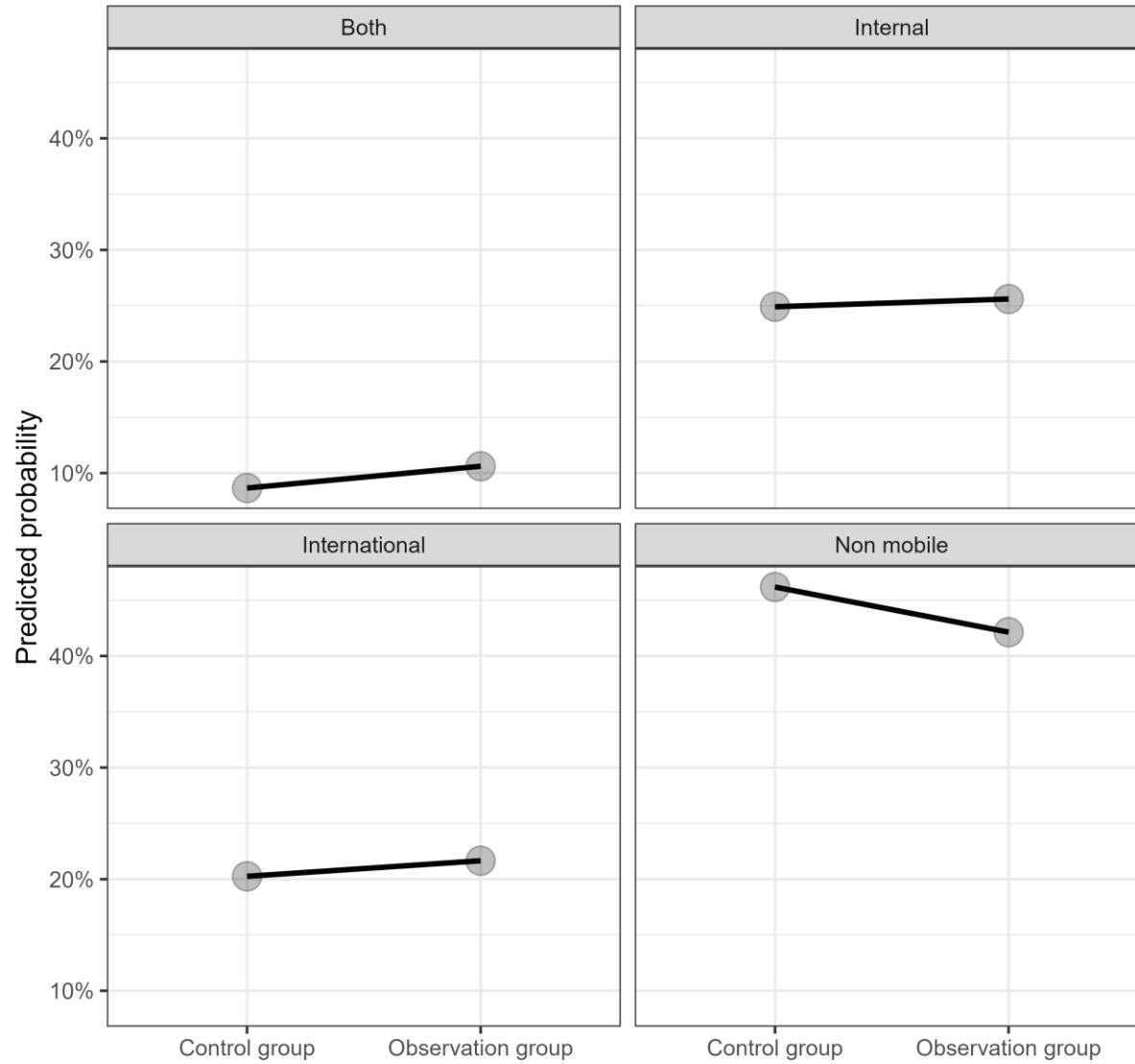
RESULTS



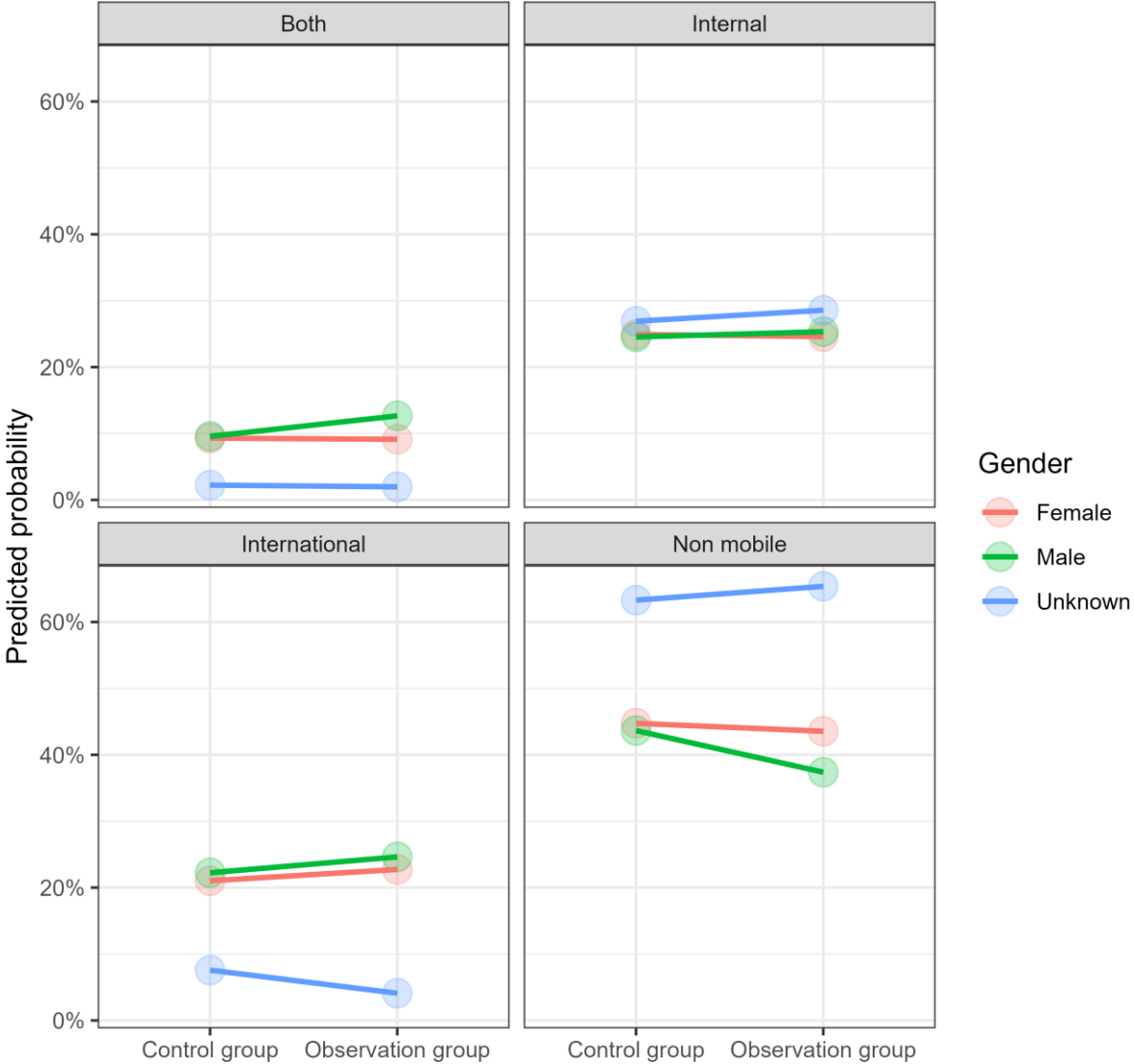
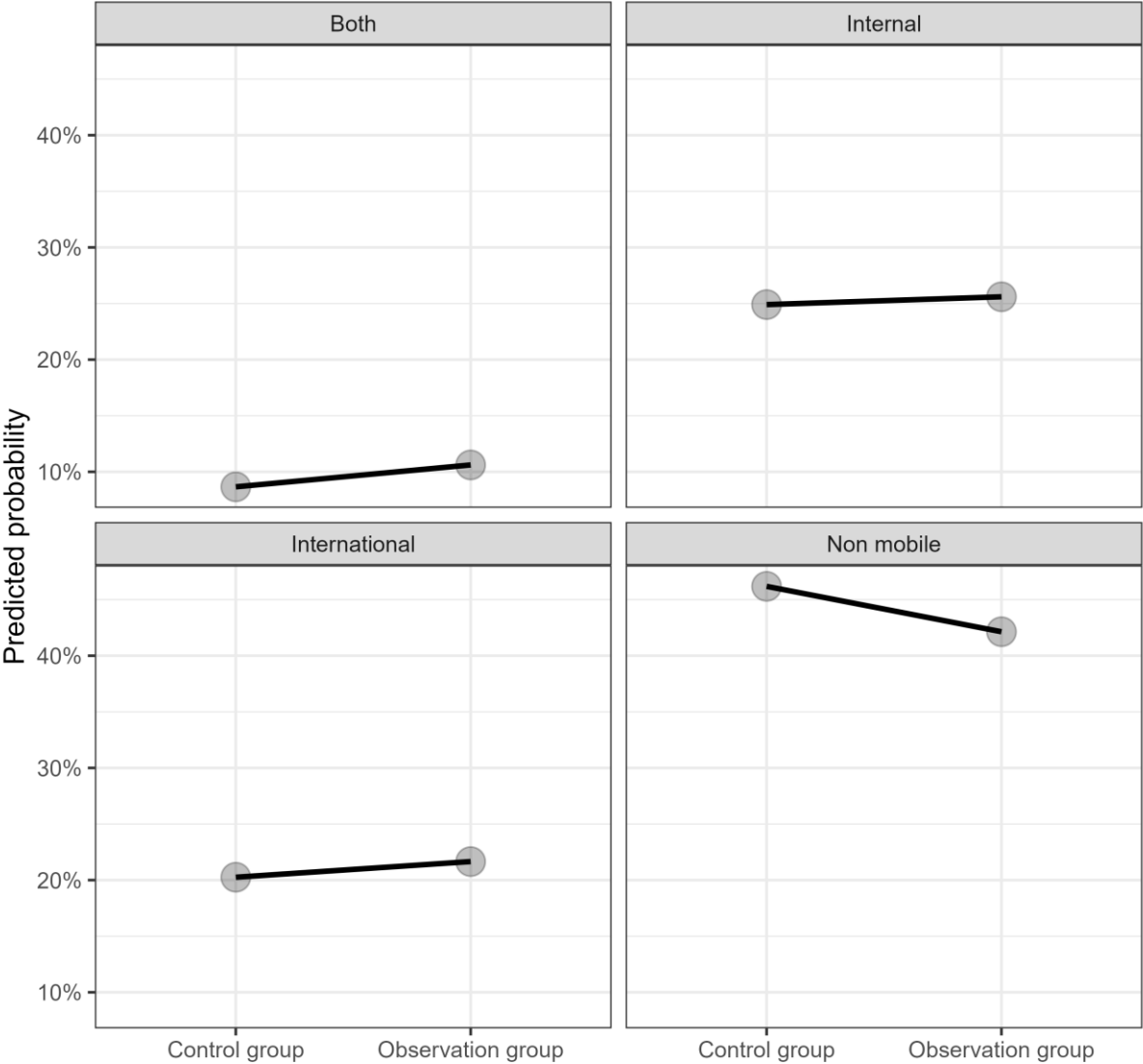
RESULTS



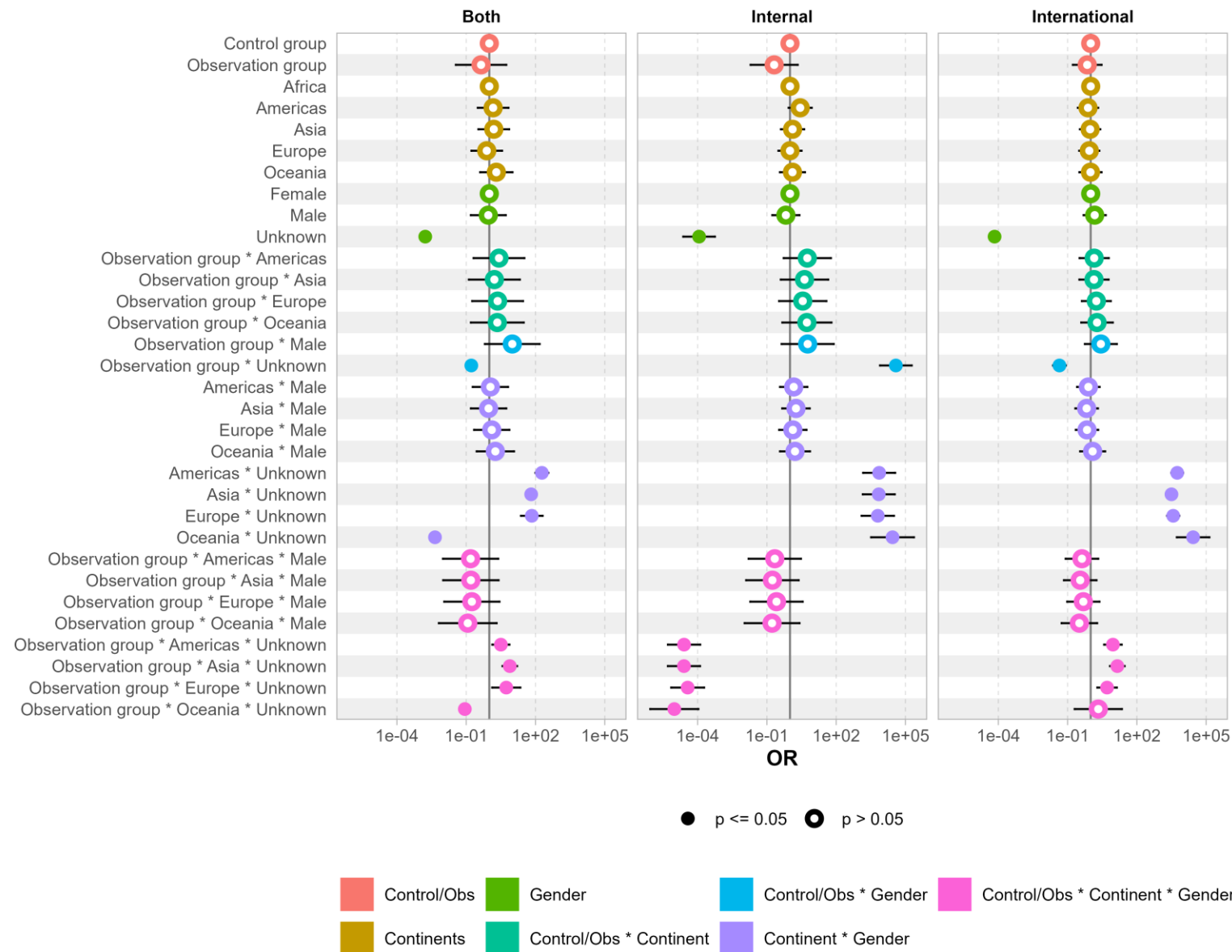
RESULTS



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RESULTS



CONCLUSIONS

Potentially talented individuals have a higher predicted probability to be mobile, more so internationally or in both types of mobility (less probability of immobility)

Women have a lower predicted probability to be mobile, more so for international mobility, higher probability of immobility

LIMITATIONS

- Scopus' coverage is dominated by English language from authors in WEIRD (Western, Educated, Industrial, Democratic) countries
- Higher mobility may be a privilege in the early career.
- Specific groups such as those with fewer caring responsibilities or visa restrictions could have better access to this privilege (Derrick et al, 2022; Zhao et al, 2023).
- While bibliometric data cannot (and should not) be the sole criterion for hiring decisions, they can aid selection committees to identify potentially talented individuals



THANK YOU FOR
YOUR ATTENTION!

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ADDITIONAL SLIDE



	<i>Categories of dependent variable: mobility</i>		
	Both	Internal	International
Observation group	0.295*** (0.042)	0.119*** (0.030)	0.158*** (0.032)
Reference category (control group)	-1.674*** (0.032)	-0.618*** (0.021)	-0.824*** (0.023)
AIC ¹		71666.17	
BIC ²		71715.69	
Log Likelihood		-35827.08	
N ³		28406	

Notes:

¹Akaike information criterion

²Bayesian information criterion

³Number of observations

Significance levels:
*p<0.1; **p<0.05; ***p<0.01